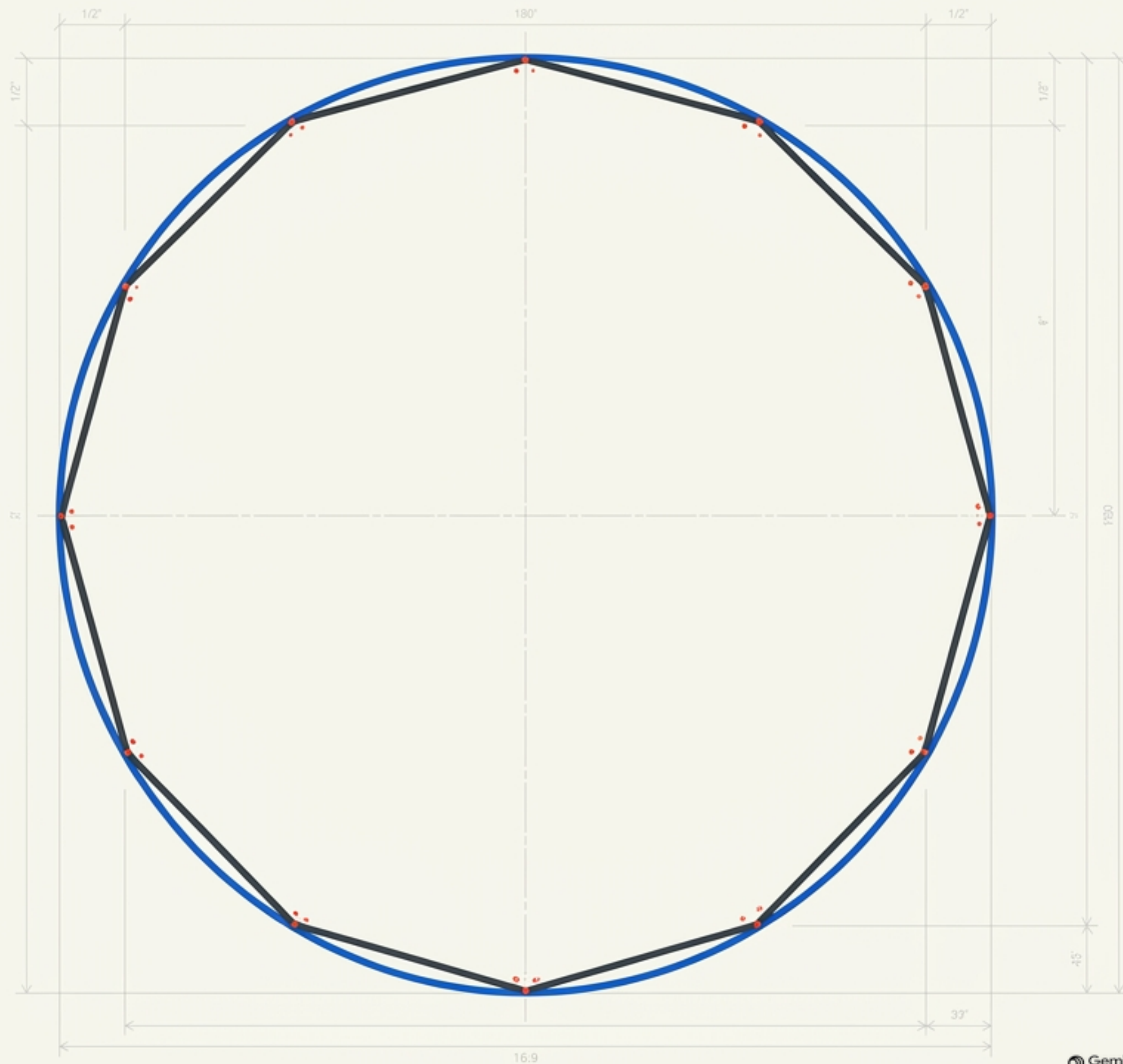
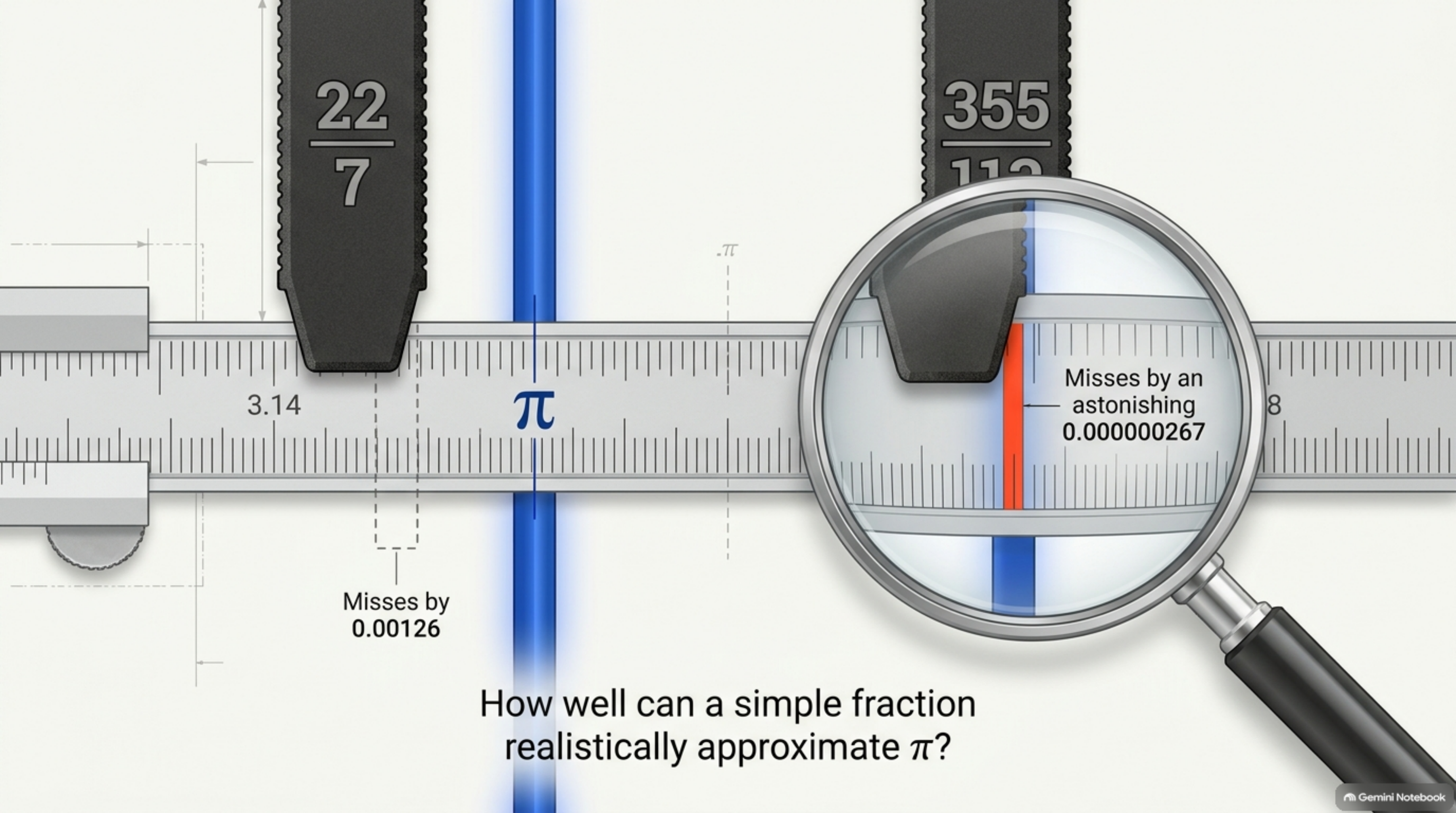


How an AI-generated proof solved the ultimate fraction problem.

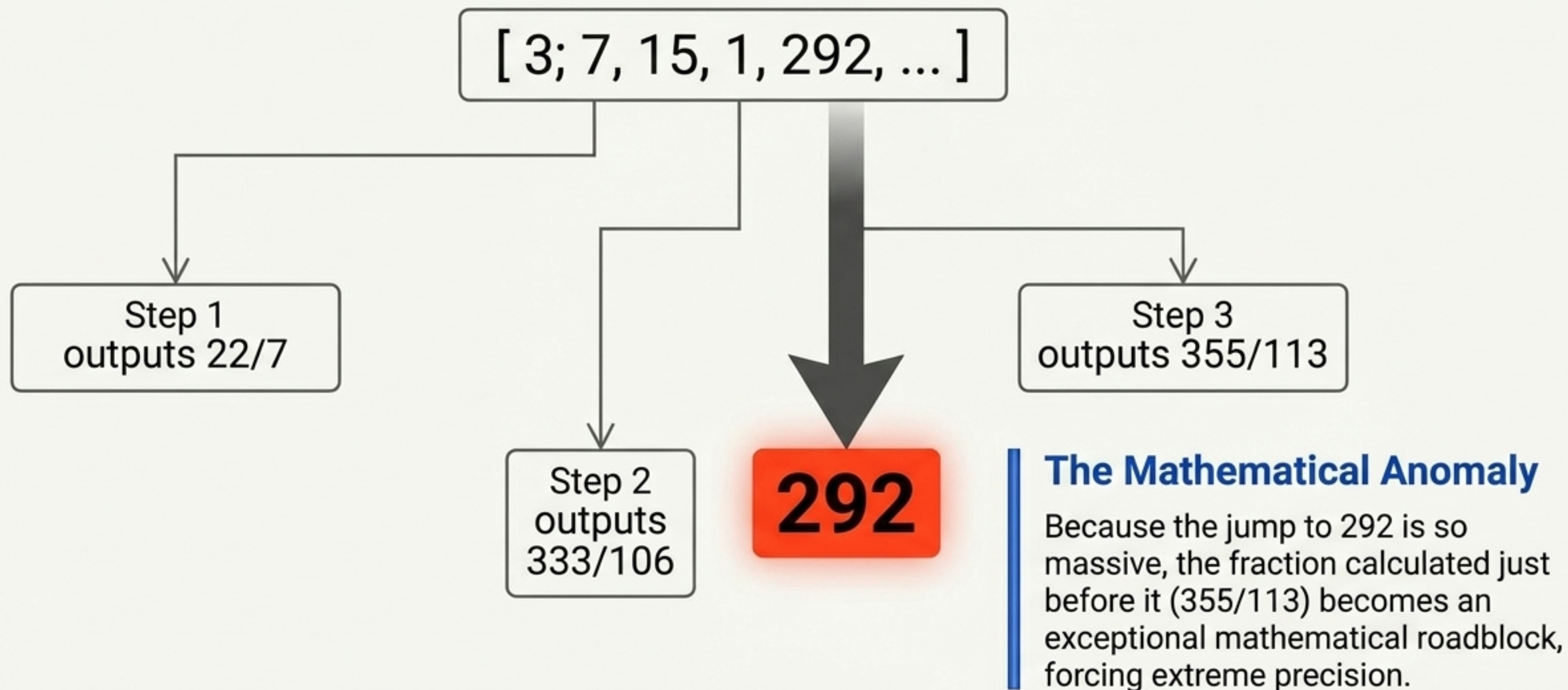
How an AI-generated proof solved the ultimate fraction problem.



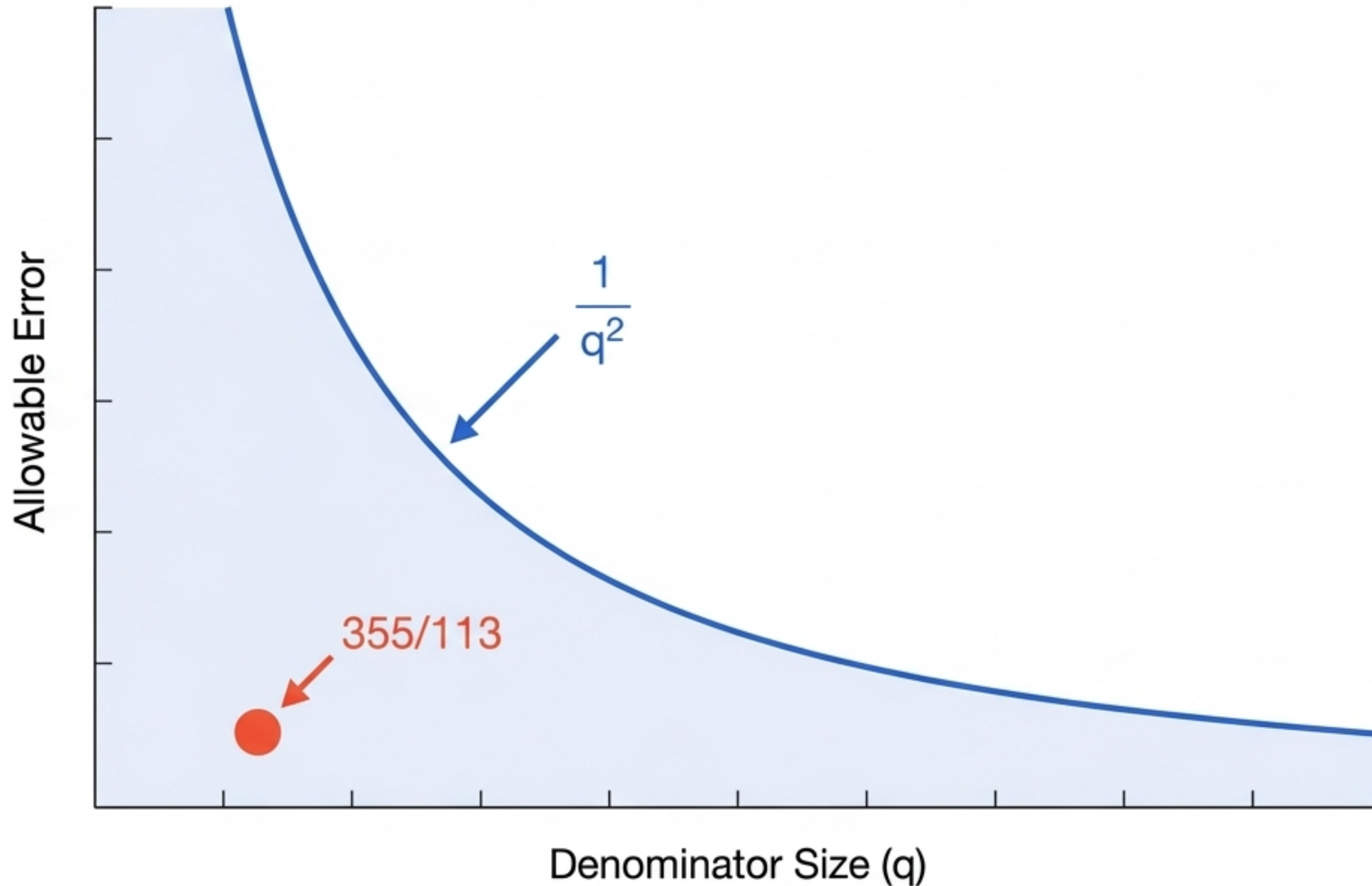


How well can a simple fraction realistically approximate π ?

The Engine of Perfect Approximations



Dirichlet's Universal Speed Limit (1842)



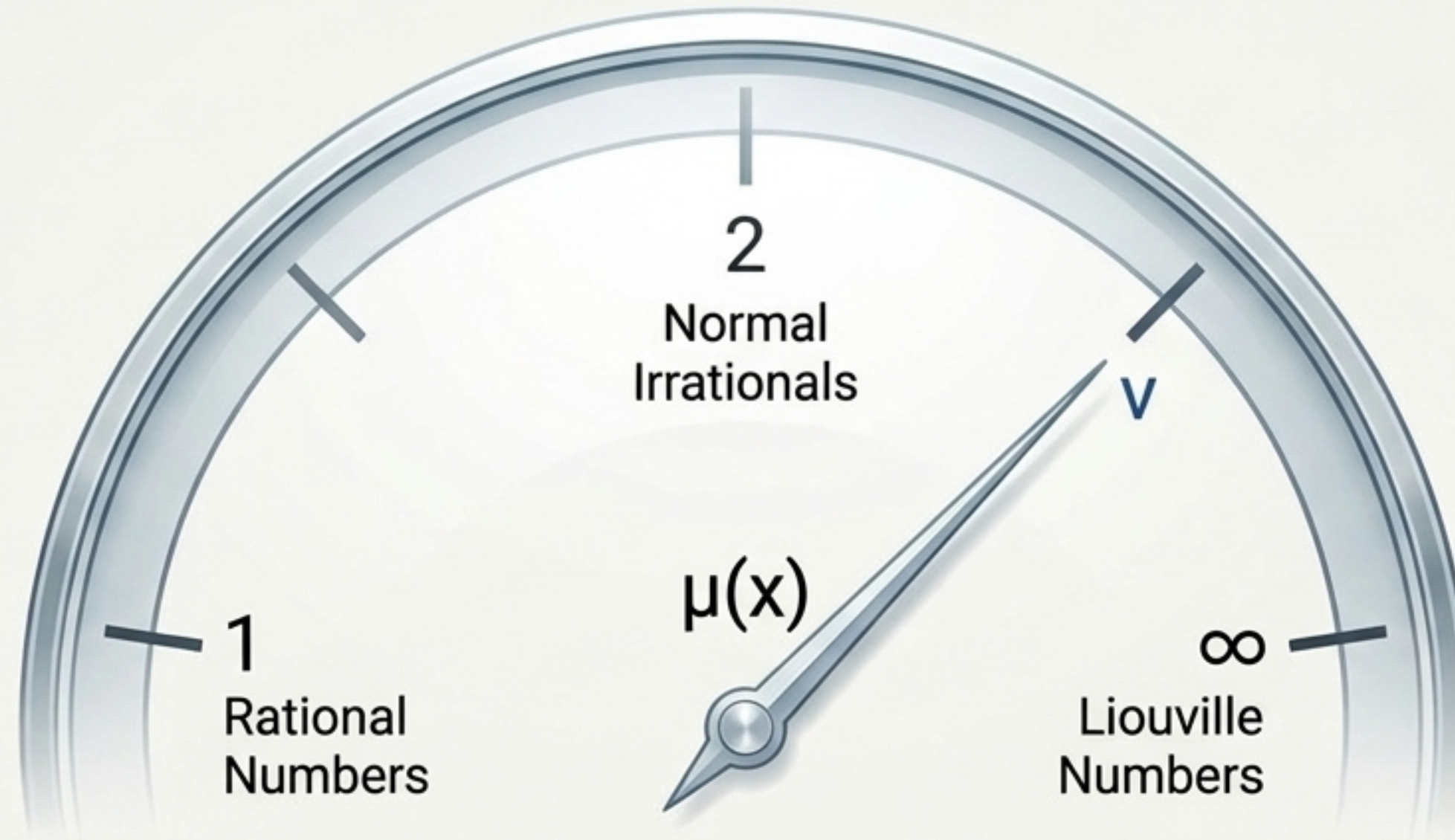
The Law

For any irrational number, you can always find infinitely many fractions where the error is strictly less than $1/q^2$.

The Paradox

$355/113$ has an error of 0.000000267. It beats the universal rule by a massive margin.

Scoring Approximations: The Irrationality Exponent



The Core Question: What is the largest power v where the error stays below $1/q^v$ for an infinite number of fractions?

The Rule: A higher exponent (μ) means the number is fundamentally easier to capture with simple fractions.

The Pi Predicament

Algebraic Numbers

Examples: $\sqrt{2}$, Golden Ratio

Solutions to simple polynomial equations.

The Law: Roth's Theorem (1955) proved all irrational algebraic numbers have an exact, rigid limit of $\mu = 2$.

Transcendental Numbers

Examples: e , π

The wild numbers that evade simple equations.

The Mystery: Roth's Theorem does not apply. The exponent is entirely unknown.

Because π is transcendental, mathematicians could not prove if its μ was 2, 2.5, or 40.

The Century-Long Race to the Ceiling

1953: Mahler sets the initial bound at 42

1993: Hata breaks into single digits at 8.016

2008: Salikhov tightens the gap to 7.606

2020: Zeilberger & Zudilin
reach 7.103

Will the ceiling keep
dropping until it hits
the absolute floor?

The Ultimate Floor: 2.0

The 2026 Breakthrough

```
proof proofchecking matrix {
  proof var v as μ = 2emoetyn
  proof var w as m with ms v)
  proof {anse} Main {
    trant=In = err * 2;
    vat m = agosxk de s: satnax (seinoatlin)
    matit: opixy^x <.2, α=0.0)
    return "llaci(maimser)
  }

proof script matCommandNi&ayt {
  proof simpnsCRode {
    vna1 μ kt => the+namsema:{{}}
    matti : 'tradl {}
    wininz-zam.fondern#kaa{[1']}
    redorn Proof inin[nix (&trdovt&l] as
    n^k4{x?} {
      true 'ind] :
      ume'bpsisitwrok-matit @
```

$\mu(\pi)$

```
proof var m = en
prior at Lean 4 language
denominator for List nnormanty)
var m = Q(π|n = Quiln()
vat om = R
matit denominato-runctions) {
  Using top na n: 2, there is hard limit Q2
  for q = π (denominator); massan ([] of v < S sn) [||]
  proof (| (coin[m, faeq =
    Zeilberger&Zoirxorroiv4Q(v2, puifin1, ftrd))1
  evren 2u,3n?
  return
}
Q=ooten: (For any_exponent > 2), there is a 1/q^v)

proof array&K&ungroodTru
proof binmas&Eode {
  mtosen k& = the+exponent{3}
  matir : 'ssm' {}
  una1 seed = angxrjanos[-mirment(u, , %
  matti {
    if (neaninvdnn k == 0 \
      return ame&assen.antis-k, 1 {3})
    } else
```

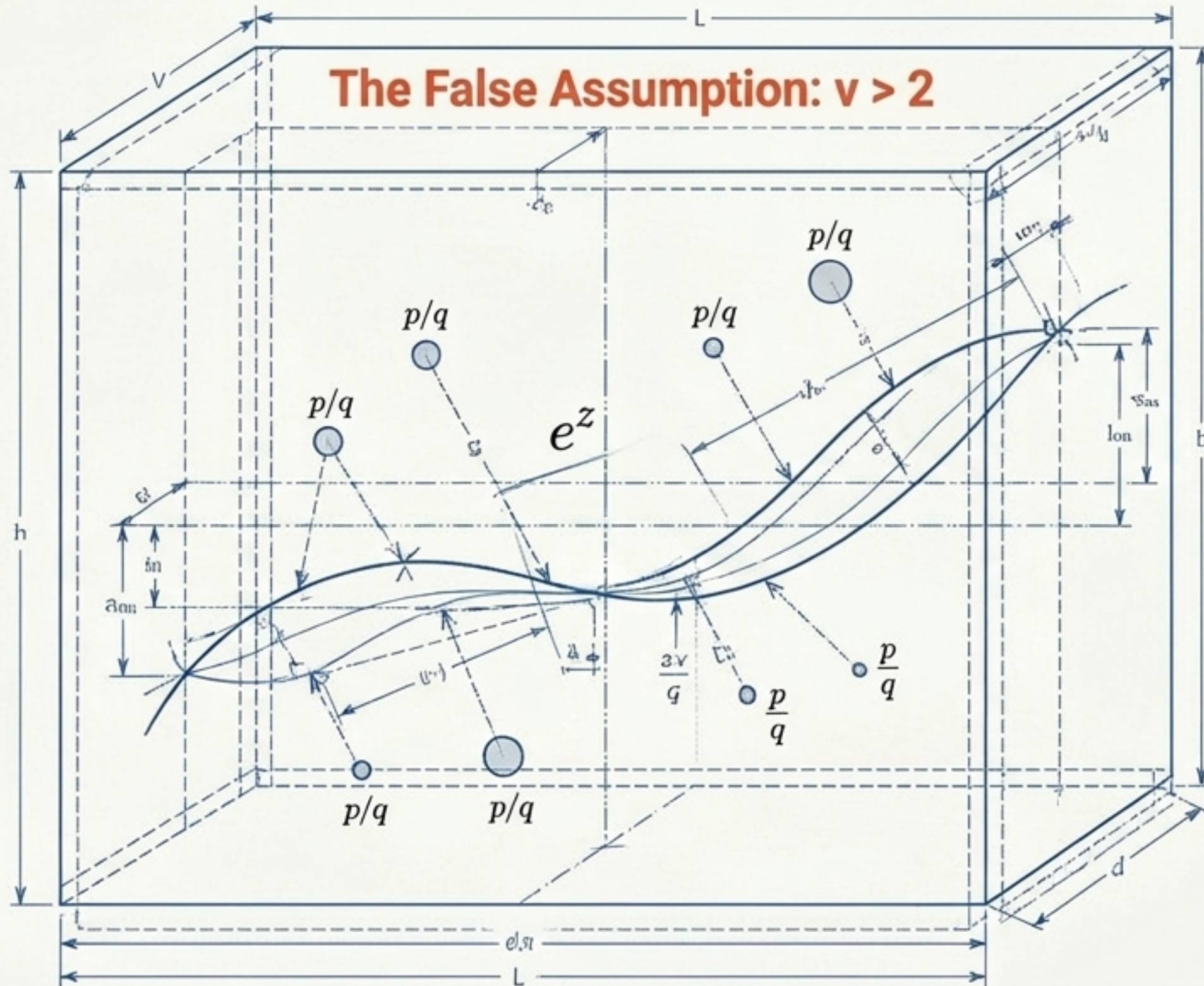
$= 2$

An AI-generated preprint shatters the record, rigorously verified by the Lean 4 software system.

The Main Theorem: For any exponent $v > 2$, there is a hard limit $Q(v)$. Once your denominator passes that limit, the error is always greater than $1/q^v$.

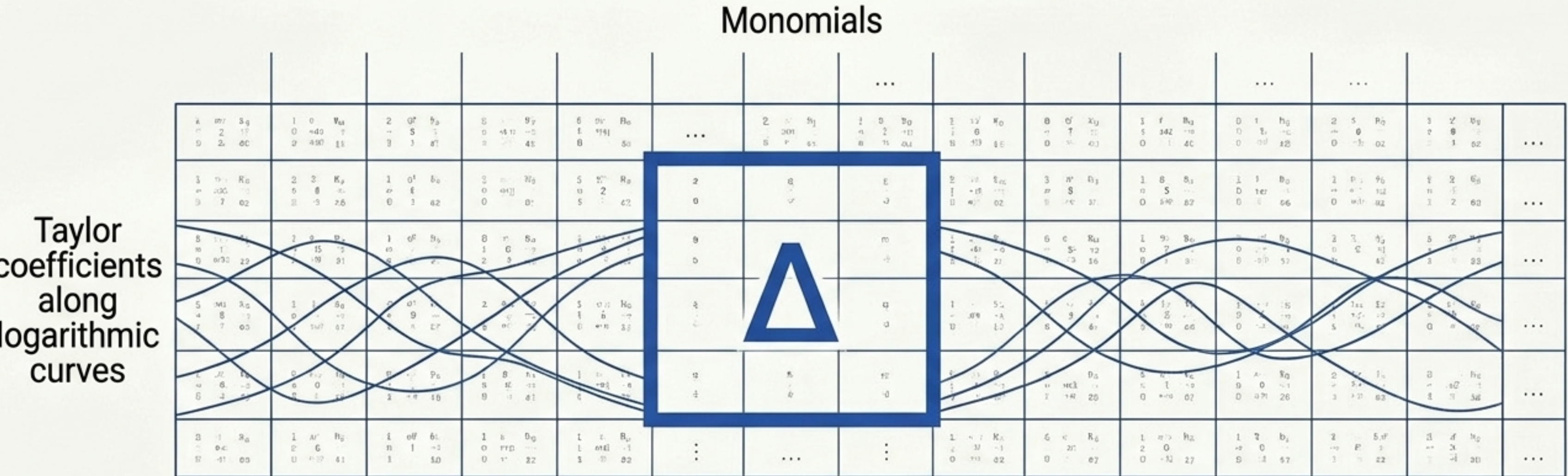
The Meaning: Fractions can only be 'so good' at approximating π , and mathematically no better.

How the Proof Works: Setting the Trap



1. **The Setup:** Assume the opposite. Assume there are infinite super-fractions with microscopic errors.
2. **The Selection:** Pick several of these super-fractions of vastly different sizes.
3. **The Mimicry:** Use these fractions to build “almost-periods” ($2i \cdot j \cdot p/q$) designed to closely mimic the exact natural periods of the exponential function.

How the Proof Works: The Independent Grid



1. The almost-periods are fed into a massive computational table.

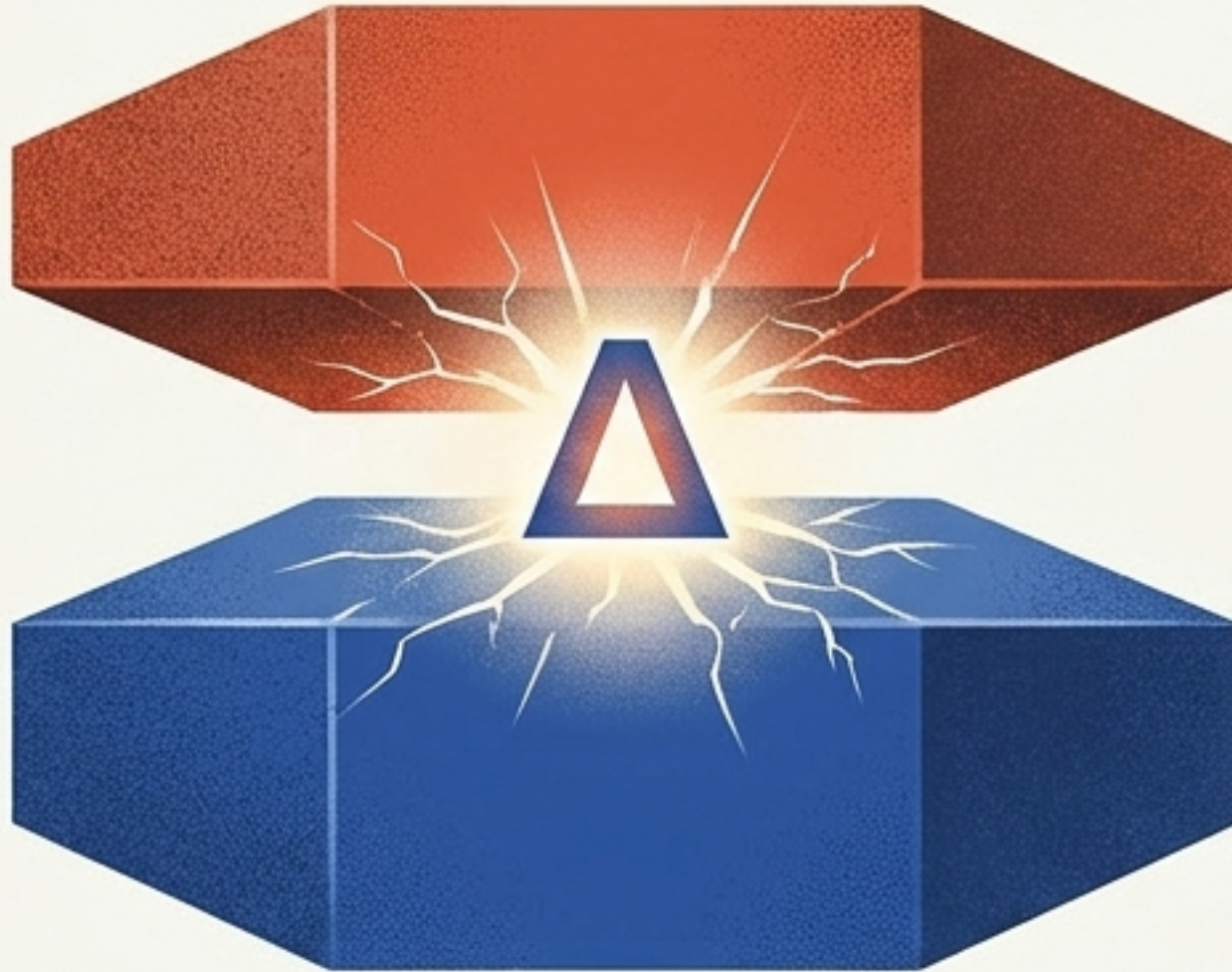
2. Using algebraic geometry, the AI proved the rows do not overlap—they are completely independent.

3. Because the rows are independent, a square block yields a **key signature**: a **determinant (Δ)** guaranteed **NOT** to be zero.

How the Proof Works: The Contradiction Crash

Upper Bound: Geometry

By sampling a smooth Vandermonde-like function, the rows carry the powers of our assumed tiny errors, forcing Δ to be microscopically small.



Lower Bound: Arithmetic

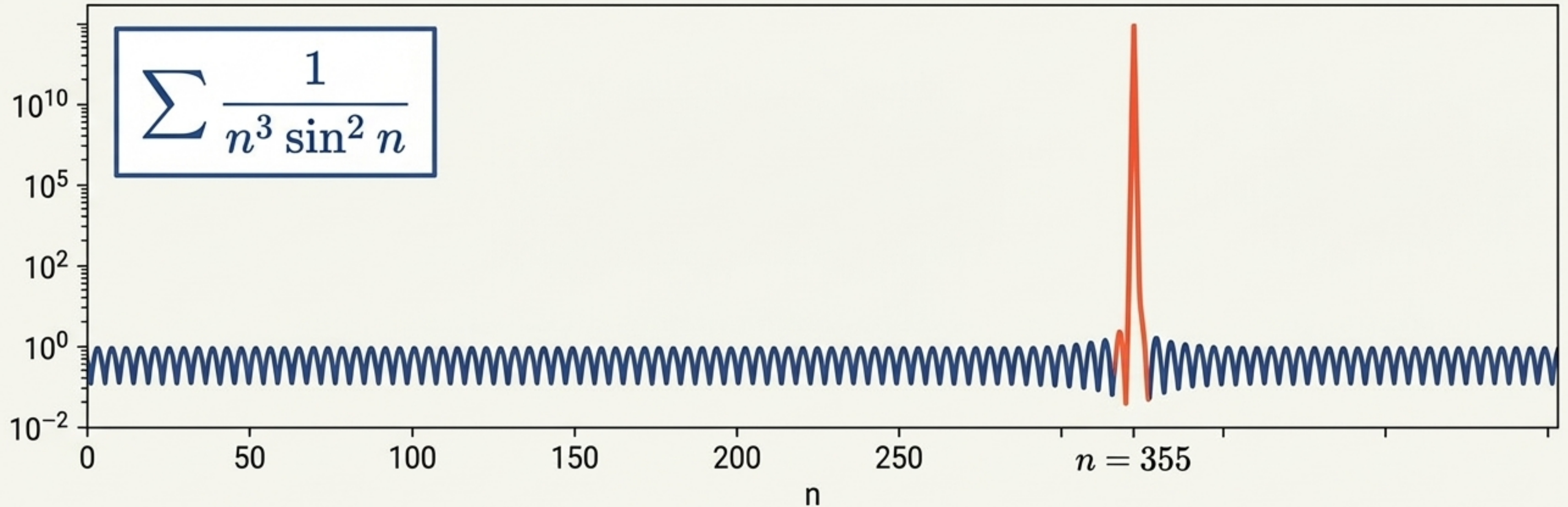
Δ is related to a nonzero Gaussian integer. It has a hard floor and can only be so small.

The Shattered Reality

For $v > 2$, the upper bound is forced below the lower bound. This is a mathematical impossibility.

The initial assumption shatters, proving $\mu = 2$.

The Ripple Effect: Solving the Flint-Hills Series



This infamous infinite sum relies entirely on how close integers get to multiples of π .

At $n = 355$, the fraction $355/113$ makes the sine term incredibly tiny, causing a single massive spike equal to ~ 24.6 .

Because we now know $\mu = 2$, we have proof that these "super-spikes" are mathematically rare enough. The series safely converges.

The Frontiers We Haven't Crossed

Uncomputable Limits

The proof guarantees a boundary $Q(\nu)$ exists where fractions stop being "super", but the AI's logic cannot tell us exactly what that physical number is.

Bounded Partial Quotients

Looking back at the massive 292 in π 's continued fraction, we still do not know if these numbers eventually hit a maximum ceiling, or if they grow infinitely larger forever.

The Architecture of the Discovery

The Question

How perfectly can simple fractions approximate the transcendental nature of π ?

The New Truth

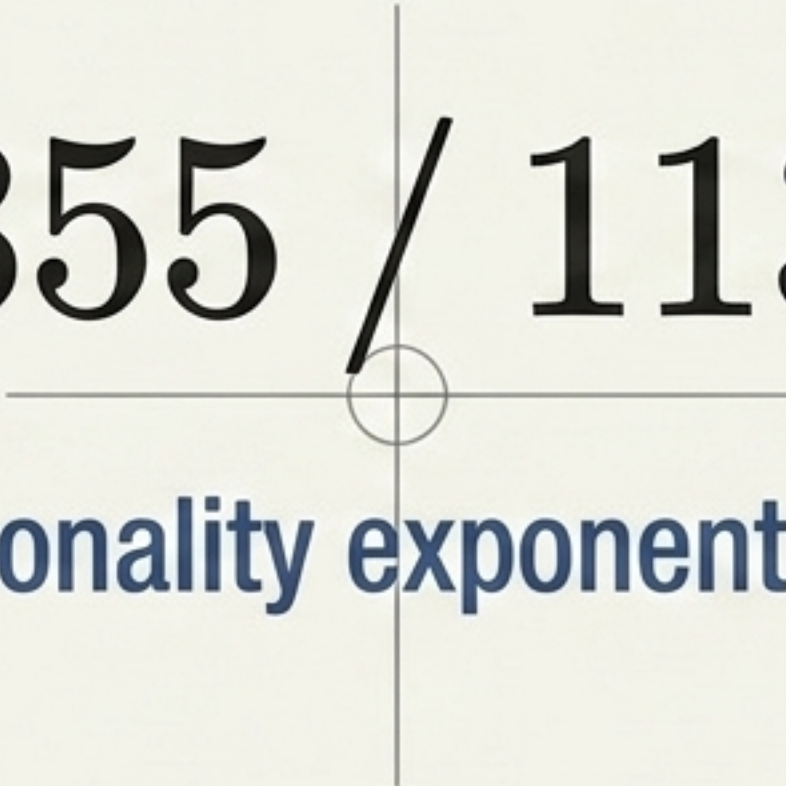
$\mu(\pi) = 2$. The absolute theoretical limit for irrational numbers, shattering the old ceiling of 7.103.

The Mechanism

Forcing an impossible mathematical contradiction between the geometry of Taylor matrices and the arithmetic of Gaussian integers.

The Impact

Proving the definitive convergence of the infamous Flint-Hills trigonometric series.


$$355 / 113$$

The irrationality exponent of π is 2.